Finals live demo: script and fallback

About 9 minutes, then Q&A (finals-judge-qa.md): 2 minutes of slides on how we decided, which the organisers asked every team to cover before the solution, then a 7-minute live demo whose centrepiece is a live call with JEV-1240 (3:00 of the 7). If the slot is shorter, drop the demo rows marked *(optional)*; never drop Part 1. Every customer and value below is pinned by tests/test_demo_customers.py and checked in a real browser by scripts/e2e_finals.mjs. Run both on the demo laptop the morning of the finals.

Roles: one driver clicks (the laptop), one narrator speaks; swap at the call so the driver plays the agent. Gemini stays off on stage; note extraction uses the local rules, which give the same result for the note below.

Before going on stage

1. Start the backend with a clean review store, so Supervisor review shows only today's items:

COPILOT_STATE=data/finals-demo-reviews.sqlite3 COPILOT_GEMINI_ENABLED=0 .venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8000

2. Start the frontend as a production build (no dev-mode overlay, faster pages):

cd frontend && npm run build && npm run start

3. Make sure data/plans-report.json exists (.venv/bin/python scripts/batch_plans.py, about 20 seconds), so the Overview shows the next-step chart.

4. Open <http://127.0.0.1:3000>, sign in as supervisor / supervisor-demo, and open each page once, plus /call/JEV-1240, so the first load is warm. Reload /call/JEV-1240 just before going on, so the call starts empty.

5. Rehearse the Overview scroll: What's at stake is a scrolling sequence (one big number at a time, with three dots under it showing which). Scrolling too fast skips the $18.0M figure the opening line depends on; scroll back up if that happens.

6. Keep the call note below in a text file to paste. Keep a second browser profile signed in as agent / agent-demo for the access question.

7. Fallback: the screenshots in data/finals-shots/ (desktop and phone, including call-JEV-1240.png) and the recorded run. If the live app fails, switch to them without apologising and keep the same narration.

Call note to paste at 4:10:

Customer lost his job last month. He can pay $315.86 on October 15. Asked us to call back next week.

Part 1: Why we built it this way (slides, about 2 minutes)

The organisers asked for the backend architecture, the reason for each model and LLM choice, and the decision process before the solution. Say the reason first, then the choice.

Slide	Say
2. The problem	"Before every call an agent has to check eight things: legal hours in the customer's time zone, holidays, a 3-in-7-days cap, insolvency, representatives, disputes, vulnerability, hardship before payment. Get one wrong and it's a breach."
3. What we found	"We ran those checks over all 96,000 open cases: 760 contacted more than allowed this week, 1,040 contacts outside legal hours, 18,758 customers in hardship with no plan."
4. How we decided	"So before writing code we set five constraints: compliance first, explainable, customer text stays inside the bank, works on the next data release, and every number measured. Then one rule for every job: if policy states the answer, rules; if it's a pattern in text with labelled examples, a small model; if it's language we can check against a source, an LLM that proposes and the app verifies; otherwise a person decides."

Slide	Say
5. Backend architecture	"Data is prepared once into DuckDB, a columnar database in one file: a 30-day scan of 2.7 million contacts takes a few milliseconds, and the app only reads it. FastAPI checks the role on every request and calls one of three engines; every answer carries its source record, policy section or SQL. Decisions and the audit log go to a separate SQLite file, hash-chained so tampering shows. Gemini sits behind the API and is optional."
6. Why this model, why this LLM	"We didn't assume an LLM would be better; we measured it. On the same 126 held-out notes, Gemini found the same 38 hardship notes but raised 4 false alarms. Three were notes tagged 'cust vulnerable - financial hardship', which our policy treats as vulnerability, not hardship. It took 1.4 seconds a note instead of under a millisecond, and it would send every note out of the bank. So the small model flags hardship, and the LLM only proposes where a person checks. We use Gemini 3.5 Flash-Lite because it is on our free tier, returns schema-checked JSON in under 2 seconds, and 3.8 Flash was overloaded the day we tested."

Part 2: Live demo (about 7 minutes)

Time	Do	Say
0:00	Overview: scroll down to What's at stake and keep scrolling slowly; the big number changes from $69.8M to $18.0M to $338.6K as you go, so pause on each	"This is debt collection: customers who already owe the bank money. $69.8 million is exposed on cases about to be written off, and $18 million is overdue from 18,758 customers in hardship who have no plan."
0:25	Point at the next-step chart, then scroll to Problems across all customers; open Definition and SQL on 760	"Every night we plan every open case: 93,320 customers in 21 seconds. Before every call an agent has to check eight things. When nobody does: 760 cases contacted more than allowed this week, 1,040 calls outside legal hours. Every number is one query, shown here."
1:00	Work queue (agent AG-7514), point at the reason column	"One agent's day, in a fixed order, not a model score. Each row says why it is there and when we may call next."
1:20	Ctrl+K, type SAV-6847, open the customer	"Ctrl+K gets to any customer."
1:30	Customer page, History & contact tab: the calendar	"When can we call her? Monday and Tuesday are blocked: three attempts in seven days. Wednesday is the National Day for Truth and Reconciliation. So Thursday, 7 am to 9 pm her time. Holidays are computed for any year, not typed in."
1:55	*(optional)* Scroll the timeline	"One history across the source systems: calls, texts, emails, promises, notes, hardship offers and cases. A contact made outside the rules is marked in red; she has one."
2:10	Ctrl+K, JEV-1240 → Call: Jérôme Vallée	"Now a live call. Jérôme has an auto loan, 86 days past due, and prefers French; the script tells the agent to offer French."
2:25	Who answered? Customer; tick Recording notice and Verify identity	"Nothing about the debt is shown until the recording notice is read and identity is verified."
2:45	Tick Hardship; point at the Live plan and Hardship options	"He says he lost his job. The plan changes immediately: offer a hardship programme before any payment request. These are his real numbers: $631.72 a month drops to $315.86 for three months, which the agent can approve. A term extension is $578.71, but needs a hardship specialist. Payment deferral is not offered, and it says why: 86 days is over the 60-day limit."
3:20	Choose on the 3-month reduced payment	"He takes the reduced payment."
3:30	Income $2,400, expenses $1,900; promise $315.86 on 15 Oct	"Is the promise affordable? $315.86 is 63% of what he has left, so it warns the agent. It is within 21 days, so the date is accepted."
3:55	*(optional)* Change the date to 30 Nov, show the block, set it back to 15 Oct	"Past 21 days, it is blocked and says the last allowed day."
4:10	Call notes: paste the note, Extract from notes	"Agents type fast. Extract proposes outcomes, each with the exact sentence it came from. Nothing changes until the agent accepts. With Gemini on, a proposal whose quote is not in the note is thrown away."
4:35	Go to wrap-up: the generated Call note for the record, outcome codes, follow-ups	"The note is written for the agent with the policy cited, the codes are set, and the one reminder before the 15th is scheduled. It counts toward the weekly cap."
4:55	Submit for supervisor review	"It goes to a supervisor; nothing is sent to the customer."
5:05	Ctrl+K, HAC-4447 → Call	"And when the agent must not call: consumer proposal. The call page is blocked and the controls are off. Contact goes through the trustee."

Time	Do	Say
5:25	Supervisor review: open the call note, Approve with a reason	"The supervisor sees the call note and outcome codes, and approves, edits or rejects with a reason. Every action is in an append-only audit."
5:55	*(optional)* Ask Collections: "Which province has the highest promise-kept rate in August 2026?"	"Judges' questions: a plain-English answer, then the table and the SQL. If we can't answer correctly, it asks for clarification instead of giving a wrong number."
6:25	Close on Overview	"It works on the next data release without code changes: dates, holidays, caps and policy versions all come from the data, and we test that with a fake 2027 release. Rules decide, people approve, and every step cites its policy. That is what makes it safe to put in front of agents."
7:00	End	

Expected values (snapshot 28 September 2026)

Where	Value
Overview, What's at stake	$69,768,875.93 exposure (expected-loss estimate $28.8M, overdue $8.1M); $18,032,649.80 overdue, 18,758 customers; $338,605.98 on 348,411 contacts
Overview, next steps (batch)	93,320 customers in 21.1 s: payment ask 39,834; promise 19,517; hardship 18,006; restriction 8,892; representative 5,176; vulnerability 1,895
Overview tiles	18,758 customers; 760 cases; 1,040 contacts; 4,100 cases; 7,052 promises; 729 cases; 3 cases
Queue AG-7514	296 open cases: 18 promise due soon, 74 hardship with no plan, 189 standard, 15 on hold
SAV-6847 calendar	Mon 28 and Tue 29 Sep: cap (3 of 3); Wed 30 Sep: holiday; Thu 1 Oct 07:00-21:00 America/Toronto
JEV-1240 call	HP-LOAN-RED3 $631.72 → $315.86 for 3 months, $947.58 moved to the end, approver agent; HP-LOAN-EXT $578.71 over 74 months, hardship specialist; HP-LOAN-SKIP not eligible (86 days past due, limit 60)
JEV-1240 promise	$315.86 on Thu 15 Oct: allowed; 63% of $500 left each month (warning); 30 Nov: blocked, "within 21 days (by Mon 19 Oct)"
JEV-1240 extract	Hardship (job loss), promise $315.86 on 2026-10-15, callback; each with its sentence
JEV-1240 wrap-up	Codes RPC, HARDSHIP_STATED, PROGRAMME_AGREED, PTP; follow-up: one reminder Tue 29 Sep from 07:00 their time
HAC-4447	"Contact blocked"; consumer proposal filed 2026-09-20; POL-COLL-001 §7.3
Ask, province ranking	Yukon highest at 40.9% (small sample: 22 promises); Nova Scotia highest among provinces with 30+ promises